Criteria for Mild cognitive impairment (MCI) - Insight46 phase 1

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Contents

Document details 2

Mild Cognitive Impairment 2

Data cleaning 3

Variables in this document 3

Document details

Categories of variables	Criteria for Mild cognitive impairment (MCI)
Summary of work undertaken	1. Classification of MCI carried out according to methods and criteria detailed below 2. Clinical diagnoses agreed by consensus – see further details below 3. Any missing data coded as -99.
Names of source variables	Source, variables, are, from, several, datasets, names, in, those, datasets, are, noted, below, in, brackets, see, the, relevant, datasheets.), From, Insight46_mmse_standardneuropsychology_cleaned_final_20190723, Logical, Memory, delayed, recall, score, Digit-Symbol, Substitution, score, From**, about, participants, concern, about, their, cognition, doctor)**, From, Insight46_ad8_cleaned_final_20190830, score
Output variables	low_logical_memory_i46p1, low_digit_symbol_i46p1, informant_concern_i46p1, participant_concern_i46p1, dementia_consensus_i46p1, mci_i46p1
Papers using these variables	1. Keuss et al. (2019) Incidental findings on brain imaging and blood tests: results from the first phase of Insight 46, a prospective observational sub-study of the 1946 British birth cohort, BMJ Open, in press 2. Lu et al. (2019) Cognition at age 70: life course predictors and associations with brain pathologies, Neurology, in press

Mild Cognitive Impairment

Criteria for Mild Cognitive Impairment (MCI) were agreed based on published criteria (Petersen, R. C. et al. (2013) 'Mild cognitive impairment due to Alzheimer disease in the community.', Annals of neurology, 74(2), pp. 199–208. doi: 10.1002/ana.23931). The principle of these criteria is that they require an absence of dementia AND evidence of objective cognitive impairment on one or more neuropsychological tests AND subjective cognitive concerns reported by the participant or an informant. For Insight 46 these criteria were operationalised as detailed below. The relevant variable names in this dataset are annotated in blue.

1. No clinical evidence of dementia (dementia_consensus = no).
2. **AND** participant concern regarding their cognition (memory or cognitive difficulties more than other people the same age, or if they felt they would seek medical attention regarding their difficulties) (participant_concern = yes) and/or informant concern regarding the participant's cognition (AD8 score ≥ 2) (informant_concern = yes)
3. **AND** objective evidence of either an amnestic (Logical Memory delayed recall ≥ 1.5 SD below the mean) (low_logical_memory = yes) and/or non-amnestic deficit (Digit-Symbol Substitution score ≥ 1.5 SD below the mean) (low_digit_symbol = yes). These cognitive tests were chosen for defining a cognitive deficit on the basis of their normal distribution across the entire cohort.

The dementia_consensus variable was agreed by consensus of Insight 46 clinicians (Jonathan Schott, Tom Parker, Chris Lane, Sarah Buchanan, Sarah Keuss, Ashvini Keshavan) based on all available information. This included the participant's self-report of any diagnoses, informant reports, clinical observations, and radiology findings.

It should be noted that some participants meeting these criteria for MCI have other clinical conditions which may or may not explain their cognitive profiles. Further details of the assessments that underpin these criteria are provided below:

Each participant had an informant who completed the AD8 questionnaire (Galvin, J. E. et al. (2005) 'The AD8: A brief informant interview to detect dementia', Neurology, 65(4), pp. 559–564. doi: 10.1212/01.wnl.0000172958.95282.2a.) See separate cover-sheet for AD8 data **(Insight46_ad8_cleaned_final_20190830)** which contains full details of this questionnaire and its administration.

Each participant completed questions about subjective cognitive decline (SCD) – see separate cover-sheet and dataset for SCD and MyCog data which contains full details of these variables, **(Insight46_scd_mycog_cleaned_final_20190830)**. As explained above, the participant_concern variable was determined based on answering 'yes' to either or both of the following questions within the SCD questionnaire: "Do you perceive memory or cognitive difficulties more than other people the same age?" and "Would you ask a doctor about these difficulties?"

As part of the cognitive assessment, each participant completed the Wechsler Memory Scale (WMS) Logical Memory test (a test of immediate and delayed recall for details in a story) and the Digit-Symbol Substitution test from the Wechsler Adult Intelligence Scale. Full details of these tests and their administration are provided in the separate data-sheet for the neuropsychology dataset (Insight46_mmse_standardneuropsychology_cleaned_final_20190723) and in the following two papers:

1 Lane et al. (2017) Study protocol: Insight 46 – a neuroscience sub-study of the MRC National Survey of Health and Development, BMC Neurology, 17:75, doi: 10.1186/s12883-017-0846-x

2 Lu et al. (2019) Cognition at age 70: life course predictors and associations with brain pathologies, Neurology, in press

Z-scores for Digit-Symbol and Logical Memory Delayed Recall were calculated based on the mean and SD of the full Insight 46 sample (variables Wz and Ldz in the neuropsychology dataset – see relevant datasheet), and a z-score of <=-1.5 was classified as a low score indicating evidence of a cognitive deficit.

Data cleaning

One participant did not complete the Digit-Symbol test (see Insight46_mmse_standardneuropsychology_cleaned_final_20190723 and associated data-sheet) so their low_digit_symbol score was coded as -99.

Variables in this document

6 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Health and medical history [6040] › Personal history of neurological disorder [60401]		
dementia_consensus_i46p1	Does the participant have a consensus diagnosis of dementia? - Insight46 phase 1	topsheet field
informant_concern_i46p1	Does the informant have concerns regarding the participant's cognition (from AD8 questionnaire)? - Insight46 phase 1	topsheet field
low_digit_symbol_i46p1	Did this participant have a low score (<= -1.5 z-score) on the Logical Memory Delayed recall test? - Insight46 phase 1	topsheet field
low_logical_memory_i46p1	Did this participant have a low score (<= -1.5 z-score) on the Digit-symbol substitution test? - Insight46 phase 1	topsheet field
mci_i46p1	Participant meets criteria for Mild Cognitive Impairment - Insight46 phase 1	topsheet field
participant_concern_i46p1	Does the participant have concerns regarding their cognition (from subjective cognitive decline questions)? - Insight46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan